

# Advanced Algorithms: Lecture 13 Notes

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## Tail Inequalities Continued

The Markov inequality actually tells us much more than we originally let on. It is easily extended by realizing that for any function  $\phi(x)$  which is non-negative and strictly monotonically increasing,

$$P(X \geq t) = P(\phi(X) \geq \phi(t)).$$

We now have any number of ways to modify the bound, as

$$P(X \geq t) \leq \frac{E[\phi(X)]}{\phi(t)},$$

for any such  $\phi$ . Moreover, it holds for any random variable  $X$  (not just non-negative!) as we only need  $\phi(X) \geq 0$  to apply Markov.

A **Chernoff bound** is simply an application of Markov with  $\phi(t) = e^{\lambda t}$  for some  $\lambda > 0$ ;

$$P(X \geq t) \leq e^{-\lambda t} E[e^{\lambda X}].$$

We can once again use this to bound the probability that  $X$  deviates significantly from  $\mu$ :

$$P(X \geq (1 + \delta)\mu) \leq e^{-(1+\delta)\lambda\mu} E[e^{\lambda X}].$$

This is particularly useful when  $X$  is a sum of independent  $[0, 1]$  random variables  $X_1, X_2, \dots, X_N$ . Suppose  $E[X_i] = p_i$  and  $\mu = E[X]$ . Then the Chernoff bound on their sum is

$$\begin{aligned} P(X_1 + \dots + X_N \geq (1 + \delta)\mu) &\leq e^{-(1+\delta)\lambda\mu} E[e^{\lambda(X_1 + \dots + X_N)}] \\ &= e^{-(1+\delta)\lambda\mu} E[e^{\lambda X_1} e^{\lambda X_2} \dots e^{\lambda X_N}] \\ &= e^{-(1+\delta)\lambda\mu} \prod_{i=1}^N E[e^{\lambda X_i}] \\ &= e^{-(1+\delta)\lambda\mu} \prod_{i=1}^N (p_i e^{\lambda} + (1 - p_i) \cdot 1) \\ &= e^{-(1+\delta)\lambda\mu} \prod_{i=1}^N (1 + p_i(e^{\lambda} - 1)) \end{aligned}$$

The first order Taylor approximation for  $e^x$  tells us that  $1 + x \leq e^x$  for all  $x \in \mathbb{R}$ . Thus

$$\begin{aligned} \mathbb{E} [e^{\lambda X}] &\leq \prod_i e^{p_i(e^\lambda - 1)} \\ &= e^{\sum_{i=1}^N p_i(e^\lambda - 1)} \\ &= e^{(e^\lambda - 1)\mu}, \end{aligned}$$

and so for all  $\lambda \geq 0$ ,

$$\mathbb{P}(X > (1 + \delta)\mu) \leq e^{-(1+\delta)\lambda\mu} e^{(e^\lambda - 1)\mu} = e^{((e^\lambda - 1) - (1+\delta)\lambda)\mu}.$$

The value of  $\lambda$  that minimizes the right hand side above is  $\lambda = \ln(1 + \delta)$ . Plugging this in and simplifying gives us

$$\begin{aligned} \mathbb{P}(X > (1 + \delta)\mu) &\leq e^{((e^{\ln(1+\delta)} - 1) - (1+\delta)\ln(1+\delta))\mu} \\ &= e^{(\delta - (1+\delta)\ln(1+\delta))\mu} \end{aligned}$$

Now we will use the Taylor series expansion of  $\ln(1 + \delta)$  given by

$$\ln(1 + \delta) = \sum_{i \geq 1} (-1)^{i+1} \frac{\delta^i}{i}.$$

Therefore,

$$(1 + \delta) \ln(1 + \delta) = \delta + \sum_{i \geq 2} (-1)^i \delta^i \left( \frac{1}{i-1} - \frac{1}{i} \right).$$

Assuming that  $0 \leq \delta < 1$ , and using the alternating series bound, we get

$$(1 + \delta) \ln(1 + \delta) > \delta + \frac{\delta^2}{2} - \frac{\delta^3}{6} \geq \delta + \frac{\delta^2}{3}.$$

Plugging this into our original expression we obtain

$$\mathbb{P}(X > (1 + \delta)\mu) \leq e^{\frac{-\delta^2\mu}{3}} \quad (0 < \delta < 1).$$

A very similar calculation shows that:

$$\mathbb{P}(X < (1 - \delta)\mu) \leq e^{\frac{-\delta^2\mu}{2}} \quad (0 < \delta < 1).$$

**Exercise.** Let  $X \sim \text{Bin}(n, p)$ . Using Markov's inequality, find an upper bound on  $P(X \geq \alpha n)$ , where  $p < \alpha < 1$ . Evaluate the bound for  $p = \frac{1}{2}$  and  $\alpha = \frac{3}{4}$ .

**Solution.** Note that  $X$  is a nonnegative random variable and  $E[X] = np$ . Applying Markov's inequality, we obtain

$$P(X \geq \alpha n) \leq \frac{E[X]}{\alpha n} = \frac{pn}{\alpha n} = \frac{p}{\alpha}.$$

For  $p = \frac{1}{2}$  and  $\alpha = \frac{3}{4}$ , we obtain  $P(X \geq \frac{3n}{4}) \leq \frac{2}{3}$ .

On the other hand, we can use Chebyshev's inequality to obtain

$$\begin{aligned} P(X \geq \alpha n) &= P(X - np \geq \alpha n - np) \\ &\leq P(|X - np| \geq n\alpha - np) \\ &\leq \frac{\text{var}(X)}{(n\alpha - np)^2} \\ &= \frac{p(1-p)}{n(\alpha - p)^2} \end{aligned}$$

For  $p = \frac{1}{2}$  and  $\alpha = \frac{3}{4}$ , we obtain

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{4}{n}.$$

Note that Markov is better when  $n$  is small, but as  $n$  increases, Chebyshev gives us a better estimate, inversely linear in  $n$ . However, we can do much better than both approaches with Chernoff. Left as an exercise for the reader.

## More General Chernoff Inequality

Chernoff bounds may also be applied to general sums of independent, bounded random variables, regardless of their distribution; this is known as **Hoeffding's inequality**. We will not cover it in this course, but it is important to prove some bounds in learning theory, among other things.